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Editor-in-Chief
IEEE Transactions on Automatic Control

Dear Editor,

We submit for your consideration the manuscript:

Physics-Constrained Adaptive Slope for Differential Protection: A Projected-Gradient Law with Lyapunov Stability Guarantees

Control-theoretic content. This paper casts a power engineering problem — adapting the slope parameter k in percentage-restrain differential protection — as a formally stated online parameter estimation problem and provides a complete stability analysis via a quadratic Lyapunov function. The central result, Theorem 1, proves uniform ultimate boundedness (UUB) of the tracking error $\tilde{k} = k - k^*$ with an explicit, computable ultimate bound ε_∞. Corollary 1 shows that the physics-derived constraint on the feasible set (the SAMBPS veto gate $\kappa_n < 30$) is *necessary and sufficient* for the constraint set to be non-empty and the UUB guarantee to hold. These are clean, self-contained control-theoretic results that use power engineering as the motivating application but stand independently.

Novelty over prior adaptive relaying work. Prior adaptive differential relaying work (*e.g.*, Kasztenny 2000, Sachdev 1989, Phadke 1988) relies on offline tuning or MRAS frameworks proven only for LTI plants. The proposed law handles a *time-varying* unknown $k^*(t)$ under a nonlinear physics-derived constraint, and provides a closed-form ultimate bound that is verifiable at design time.

Five contributions.

1. Formal problem statement: slope as constrained time-varying parameter with a physics-derived feasible set $\mathcal{K}(\kappa_n)$.
2. Projected-gradient adaptive law (Algorithm 1) implementable at 100 ms cycle rates on embedded DSPs.
3. Lyapunov UUB theorem with explicit bound ε_∞ and convergence rate α.
4. Necessity-and-sufficiency proof linking the control-theoretic result to the model-based protection veto gate.
5. $7\times$ RMSE improvement over fixed-slope alternatives on a 33 kV mixed-IBR feeder (2400 simulation scenarios).

Fit to the journal. *IEEE Transactions on Automatic Control* has a strong tradition of Lyapunov-based adaptive control theory and its application to engineering systems. This paper makes a new connection between adaptive parameter estimation with projection and the power system protection domain, where formal stability guarantees are currently absent from the literature.

Suggested reviewers.

- Prof. Petros Ioannou (University of Southern California) — robust adaptive control, projection operators
- Prof. Romeo Ortega (Université Paris-Saclay) — Lyapunov stability, port-Hamiltonian methods
- Prof. Anna Stefanopoulou (University of Michigan) — constrained adaptive estimation
- Prof. Anuradha Annaswamy (MIT) — adaptive control theory

This manuscript is not under review elsewhere. All authors have approved the submission.

Sincerely,

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